

Reproeconomics

The Economic Impact of State-Funded Oocyte Cryopreservation

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1. The policy problem

Spain combines two facts: fertility is structurally low, and motherhood is increasingly postponed.

later motherhood $\implies$ higher biological constraint.

Elective oocyte cryopreservation (OC) is attractive because it preserves a future fertility option:

$$V_i^{OC} = \mathbb{E}\{U_i(\text{future fertility option})\} - C_i^{\text{private}}.$$

The public question is different. It asks whether public funding creates additional births:

$$W^{\text{public}} = N_{\text{add}} \cdot NFC - C_{\text{programme}}^{\text{public}}.$$

Policy distinction

A clinically successful technology is not automatically a demographically effective or fiscally viable public policy.

2. Research question

The central question is whether state-funded elective OC can be justified as a pro-natalist and fiscally sustainable intervention.

The relevant outcome is not the number of funded cycles, but the births that would not otherwise have occurred:

$$N_{add} = N_{OC\ births} \cdot \alpha.$$

Additionality

$$\alpha = \Pr(\text{birth would not have occurred without OC} \mid \text{OC-assisted birth}).$$

$$\alpha = 1$$

Every OC-assisted birth is additional; strong pro-natalist mechanism.

$$\alpha < 1$$

Part of the programme is substitution, timing, or insurance.

The empirical task is to discipline α using Spanish fertility dynamics.

3. Empirical strategy: counterfactual fertility

The demographic block estimates whether fertility after 2010 recovers relative to a pre-break counterfactual.

$$\widehat{Y}_t^0 = \widehat{\beta}_0 + \widehat{\beta}_1 t + \widehat{\delta}' X_t, \quad \widehat{\tau}_t = Y_t - \widehat{Y}_t^0.$$

The interrupted time-series specification is:

$$Y_t = \beta_0 + \beta_1 t + \beta_2 D_t + \beta_3 (t - T_0) D_t + \delta' X_t + u_t, \quad T_0 = 2010.$$

$$D_t = \begin{cases} 0, & t < T_0, \\ 1, & t \geq T_0. \end{cases}$$

Identification status

This is calibration, not causal identification: Spain had no national elective-OC programme during the estimation period.

Annual Spanish data, 1976–2022. HAC standard errors, lag = 3.

4. Reading the coefficients

The ITS coefficients separate the post-2010 shock from the subsequent adjustment path.

β_2 = post-break level shift, β_3 = post-break slope correction.

Hence the estimated gap is:

$$\hat{\tau}_t = \hat{\beta}_2 + \hat{\beta}_3(t - T_0).$$

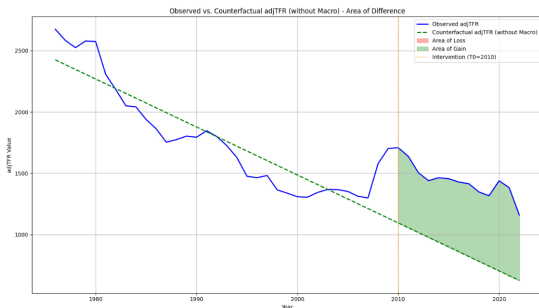
The interpretation is direct:

$\hat{\tau}_t < 0 \Rightarrow$ fertility remains below its pre-break counterfactual.

For OC policy, this matters because additionality requires more than delayed births. A programme can be clinically successful and still fail to generate enough extra births to satisfy the fiscal test.

The empirical block therefore supplies a disciplined prior on demographic additionality.

5. Aggregate fertility result



Observed adjusted total fertility versus counterfactual.

The shaded area is the fertility gap relative to the pre-2010 trajectory.

$$\hat{\beta}_2^{adjTFR} = -1399.35^{***}$$

$$\hat{\beta}_3^{adjTFR} = +45.23^{***}.$$

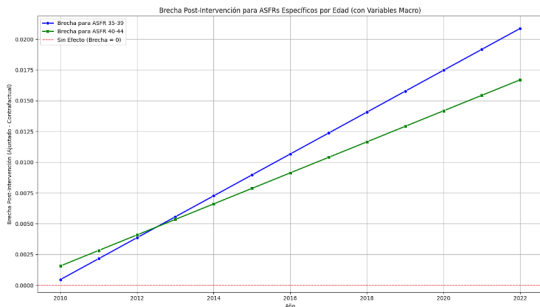
The level shift is large and negative. The post-break slope is positive, but the gap does not close:

$$\sum_{t=2010}^{2022} \hat{\tau}_t < 0.$$

Main aggregate result

Aggregate fertility stabilises partially, but does not recover the counterfactual path.

6. Age-specific fertility result



Post-break gap in fertility rates at ages 35–39 and 40–44.

The key empirical finding is asymmetric.
At ages 35–39, the gap narrows:

$$\hat{\tau}_{35-39,t} \nearrow 0.$$

At ages 40–44, the gap remains negative:

$$\hat{\tau}_{40-44,t} < 0 \quad \forall t \geq 2010.$$

This matters because OC funded at 30–35 would usually be used later:

$$30-35 \Rightarrow 38-45.$$

Main age-specific result

The evidence supports partial postponement, not complete intertemporal substitution.

7. Cost-effectiveness model

Every funded woman generates the upfront vitrification cost, even if she never uses the oocytes.

$$\mathbb{E}[C] = C_u + uC_T.$$

An additional live birth requires three filters:

$$\underbrace{u}_{\text{utilisation}} \cdot \underbrace{s}_{\text{success}} \cdot \underbrace{\alpha}_{\text{additionality}} \cdot$$

Cost per additional live birth

$$ICER = \frac{C_u + uC_T}{us\alpha}$$

The denominator is small because costs are paid upfront, but births appear only after utilisation, success, and additionality.

8. Base-case ICER

Base-case parameters:

$$C_u = \text{€}5,000, \quad C_T = \text{€}2,500, \quad u = 0.08, \quad s = 0.30, \quad \alpha = 0.50.$$

Then:

$$ICER_{base} = \frac{5,000 + 0.08(2,500)}{0.08 \cdot 0.30 \cdot 0.50} = \text{€}433,333.$$

This is the expected public cost per additional live birth, not per funded cycle.

funded cycle $\neq$ use $\neq$ successful birth $\neq$ additional birth.

The model is most sensitive to utilisation and additionality, not only to the clinical success rate.

9. Fiscal viability

The fiscal benchmark is the lifetime net fiscal contribution of an additional person.

$$NFC_{2006} = \text{€}66,709, \quad NFC_{2023} \simeq \text{€}91,391.$$

Fiscal self-sustainability requires:

$$ICER \leq NFC.$$

Fiscal result

$$\frac{NFC_{2023}}{ICER_{base}} = \frac{91,391}{433,333} = 0.21 \text{ €}/\text{€}$$

The baseline fiscal return is far below break-even.

10. Break-even utilisation

The break-even utilisation threshold is:

$$u^* \geq \frac{C_u}{s\alpha NFC - C_T}.$$

Even under a favourable scenario,

$$s = 0.35, \quad \alpha = 0.60, \quad NFC = \text{€}91,391,$$

we obtain:

$$u^* = \frac{5,000}{0.35 \cdot 0.60 \cdot 91,391 - 2,500} = \mathbf{0.30}.$$

$$0.30 \approx 4 \times 0.08.$$

Break-even condition

Break-even requires utilisation around 30%, almost four times the empirical average of 8%.

11. Conclusion

The thesis does not claim that OC is useless. It claims that its strongest public justification is distributive, not demographic or fiscal.

$$\widehat{\tau}_{adjTFR,t} < 0 \quad \text{after 2010,} \quad \widehat{\tau}_{40-44,t} < 0 \quad \text{despite partial recovery at 35-39.}$$

$$ICER_{base} = \text{€}433,333 > NFC_{2023} = \text{€}91,391.$$

Thus, under empirically plausible values of (u, s, α) ,

public OC is not fiscally self-sustaining as pro-natalist policy.

If adopted, it should be designed as an equity policy:

$$Subsidy_i = f(Y_i), \quad \frac{\partial Subsidy_i}{\partial Y_i} < 0, \quad P_{cycle}^{public} \leq \bar{P}.$$

12. Final message

Final message

OC may insure reproductive autonomy, but it should not be sold as a fertility fix.

The empirical evidence points to incomplete recuperation after postponement, while the fiscal model shows that the cost per additional birth exceeds the benchmark net fiscal contribution.